CSE402: Optimisation

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To be written...

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1 Lecture 01

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Mathematical language of this course is linear algebra (of real vector spaces) and multivariable calculus. In this lecture, we will give a brief review of linear algebra.

§1.1 Review on linear algebra

Notation. Let $a_1, a_2, \ldots, a_n$ be a finite sequence of real numbers. We will denote the sum

$$S = a_1 + a_2 + \cdots + a_n \tag{1.1}$$

by

$$S = \sum_{k=1}^{n} a_k. \tag{1.2}$$

We have the following useful identities:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \tag{1.3}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{1.4}$$

§1.1.1 Vectors

Definition 1.1.1 ($\mathbb{R}$-vector space)**.** Let V be a non-empty set. The set V equipped with two binary operations

$$+ : V \times V \to V$$

called *vector addition*, and

$$\cdot : \mathbb{R} \times V \to V$$

called *scalar multiplication* is called a **vector space over** $\mathbb{R}$ or **$\mathbb{R}$-vector space** if for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$, the following axioms hold:

1. **Closure under addition:**
$$u + v \in V \tag{1.5}$$

2. **Commutativity of addition:**
$$u + v = v + u \tag{1.6}$$

3. **Associativity of addition:**
$$(u + v) + w = u + (v + w) \tag{1.7}$$

4. **Existence of zero vector:** There exists a vector $0 \in V$ such that

$$u + 0 = u \tag{1.8}$$

5. **Existence of additive inverse:** For every $u \in V$, there exists $-u \in V$ such that

$$u + (-u) = 0 \tag{1.9}$$

6. **Closure under scalar multiplication:**

$$au \in V \tag{1.10}$$

7. **Distributivity of scalar multiplication over vector addition:**

$$a(u + v) = au + av \tag{1.11}$$

8. **Distributivity of scalar multiplication over field addition:**

$$(a + b)u = au + bu \tag{1.12}$$

9. **Associativity of scalar multiplication:**

$$a(bu) = (ab)u \tag{1.13}$$

10. **Identity element of scalar multiplication:**

$$1u = u \tag{1.14}$$

Hence, $(V, +, \cdot)$ is called a vector space over the field $\mathbb{F}$. Elements of V are called **vectors** and elements of $\mathbb{R}$ are called **scalars**.

Theorem 1.1.2
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